DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #05G: GRAND FINALE — MASTER END-TO-END AUDIT OF THE ENTIRE EQUIPMENT ECOSYSTEM (898 ITEMS, 1,370 RECIPES, WEAPONS + ARMOR + ACCESSORIES + TOOLS + PROCEDURAL + LEGENDARY + ECONOMY + CPE + MARKET + AI + COMBAT + SAVE/LOAD)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Role of Document `05G`**: **Grand Finale & Unified Architectural Synthesis** concluding the 12-document Equipment & Arsenal Deep Audit (`04A–04E` Weapons + `05A–05G` Non-Weapons, Procedural & End-to-End Ecosystem). Every cross-module seam between `economy.js`, `cpeco.js`, `external_market.js`, `sect.js`, `cognition.js`, `martial.js`, `stats.js`, `combat.js`, `visitors_clues.js`, `explore.js`, `expedition.js`, `fishing.js`, `sim.js`, and `ui.js` is audited here for first-, second-, and third-order emergent consequences.

---

0. Non-Negotiable Engine Invariants

1. **Immutable `human_core.js`**: Byte-for-byte SHA-256 lock (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Zero Render RNG**: Preserve `<script>` order in `index.html`; `UI` / `Render` (`ITEM_ART_CATALOG.json`, 9 WebP atlases, equipment tier halos) must never mutate item stats or consume `W.rng`.
3. **Closed-Loop Material & Artifact Conservation**: Every forged, traded, gifted, inherited, stolen, or salvaged equipment item must have physical provenance, deterministic Save/Load stats, and 100% autonomous NPC causality.

---

1. [EQ-SYS-01] Critical Architecture Flaw: The 4-Layer Monkey-Patched `ECO.mintGear` Onion (`economy.js` $\to$ `stats.js` $\to$ `martial.js` $\to$ `visitors_clues.js`) Corrupts Authored `source.st` Items!

Tracing the runtime call stack of `ECO.mintGear` across the 40-script boot sequence exposes a **severe four-file wrapper collision**:

1. **Layer 1 (`js/economy.js:231–247`)**: Clamps `q = clamp(q, 0.7, 1.5)` (silently capping any `> 1.50` refined/spawned quality!) and omits `listedAtDay`.
2. **Layer 2 (`js/stats.js:689–703`)**: Copies `it.effects`, `grantedSkills`, `skillModifiers`, `composableEffects`, and `charges`.
3. **Layer 3 (`js/martial.js:689`)**:

```javascript
const r = (a && a.sect && a.npcs) ? b : a; // <-- Diverges from economy.js/stats.js W-detection!
...
if (source.st && !it.st && !source.fixedStats) {
  const qf = it.q || 1;
  it.st = Object.fromEntries(Object.entries(source.st).map(([k,v]) => [k, ...]));
}
```

4. **Layer 4 (`js/visitors_clues.js:698–728`)**: `VC.mintLegendaryEquipment(W, id)` calls `ECO.mintGear(def, def.q)` **without passing `W`**, desyncing `W.sect.gearUidSeq` and `listedAtDay`!

Two Devastating Stat Corruptions Caused by Layer 3 (`martial.js:689`) vs `ST.gearStats` (`stats.js:44–92`):

- **Corruption A — `it.effects` Stat Bonuses Skipped on Mint (`Hundreds of EQ_V200_* / EQ_V215_* / EQ_V230_* / REL_V230_* Items`)**:
  - Because `martial.js:689` pre-populates `it.st` from `source.st`, when `ST.gearStats(it)` (`stats.js:45`) is called, **`if (it.st) return it.st;` immediately short-circuits on line 45**!
  - Lines `66–82` of `stats.js` (`armorReductionRatio * 40` pen, `EFX_BARRIER_SHIELD` def/mdef, `EFX_V234_*` `bonusAtkPct`/`bonusCrit`/`bonusPen`) **never execute** on any `source.st` item!

- **Corruption B — Permanent Stat Identity Destruction on Rune Socketing or Kill Evolution (`delete gear.st`)**:
  - When `ECO.socketElementalRunes` (`economy.js:1294`) or `ECO.recordWeaponKill` (`economy.js:1093`) runs **`delete gear.st; ST.gearStats(gear);`**, `ST.gearStats` (`stats.js:44–92`) checks `it.baseSt` (which **only procedural items have**—`martial.js:689` never saved `it.baseSt`!) and **never checks `ECO.itemById[it.id].st`**!
  - Result: **The authored `source.st` of `EQ_V215_*`, `EQ_V200_*`, `EQ_V230_*`, and `REL_V230_*` Mythic Relics is permanently wiped out and replaced by generic hash-rolled stats**!!

Unified Single-Authority Fix:

- Store `it.baseSt = Object.assign({}, source.st)` and `it.baseQ = 1, it.baseTier = it.tier` whenever `source.st` exists, **do NOT pre-bake `it.st` in `martial.js:689`**, and let `ST.gearStats(it)` be the **sole** deterministic builder (`baseSt/fixedStats/formula + affixes + effects + artisan`) across initial mint, catalyst refinement, rune socketing, and save/load!

---

2. `[EQ-SYS-02]` Broken Social & Economic Loops in `js/cpeco.js` (`Gift Interception`, `Missing Personal Forge`, `Double Price Penalty`, `Broken-Gear Buying`, & `HERB_POOL` Pollution)

Line-by-line inspection of `js/cpeco.js` reveals **five cross-system bugs** that choke emergent NPC behavior around equipment:

2.1 `A.donate` Intercepts Every `'gift-worthy'` Heirloom Before Mentors/Spouses Can Gift It (`cpeco.js:183–187, 238–244`)!

- `MARTIAL.classifyGearForNpc` (`martial.js:638–655`) identifies when a spare sword/robe in `n.inv.gear` is a major upgrade for the NPC's **personal disciple (`apprentice`), spouse (`คู่ครอง`), or sworn sibling**, returning `{ category: 'gift-worthy', targetNpcId: o.id }`.
- **Bug**: `personalSurplusGear` (`cpeco.js:185`) returns `cls.category === 'surplus' || cls.category === 'gift-worthy'`, and `A.donate.done` (`line 239`) dumps **all** of them straight into `SECT.EDITOR.donatePersonalGear` (the impersonal CP Shop—where poor junior disciples lack the CP to buy it!) instead of gifting it!

- **Fix**: In `A.donate.done` (or `CPE.evaluatePersonalBagGear`), when `cls.category === 'gift-worthy'` and `cls.targetNpcId` is alive in the sect (`loyalty/affection` check), invoke **`COG.giveGearGift(W, n, W.byId[cls.targetNpcId], eq)`** directly—creating authentic Master-to-Disciple and Spouse-to-Spouse equipment inheritance!

2.2 Alchemists Have `A.personalBrew`, But Blacksmiths / Talisman Scribes / Array Masters Lack `A.personalForge` (`cpeco.js:440–495`, `economy.js:297`)!

- `ECO.personalCraftQuote` (`economy.js:297`) hardcodes `if (r.outType !== 'PILL') return null;`!
- An Alchemist can brew personal pills from `n.inv.mats`, **yet a Master Blacksmith (`SPIRIT_SWORD_ARTIFICER` / `HEAVENLY_ANVIL_SMITH`) carrying personal Meteor Iron (`n.inv.mats`) can NEVER forge a personal weapon, armor, tool, or talisman for themselves or their disciple**!
- **Fix**: Extend `ECO.personalCraftQuote` / `A.personalForge` at `forge` so crafters with `FORGING` / `ARRAY` / `TALISMAN >= 25` and personal materials in `n.inv.mats` can forge personal gear once every 20 days (paying CP for any missing secondary material from `W.sect.stock`).

2.3 Double Price Subtraction & Stale `gearLen` Cache in `CPE.pickPurchase` (`cpeco.js:254, 257, 291–295`)!

1. `M.gearSuitability(n, gr, price)` (`martial.js:607`) **already** subtracts `price * thrift` from `fit.score`, yet `CPE.pickPurchase` (`line 294`) compares `(rawPower - price * thrift) > cs * 1.20 + 8` **and** `consider()` (`line 257`) subtracts `price / cp * 0.4` a second time—forcing Tier 5–6 armory upgrades to be nearly `+55%` stronger before disciples buy them (causing good gear to sit 6 days and get liquidated by `sellStaleSectGear`!). Pass `price = 0` into `M.gearSuitability(n, gr, 0)` when comparing physical power `gs > cs * multReq + flatReq`, letting `consider()` handle the CP price affordability weight once!
2. `_pickCache` (`cpeco.js:254`) checks `n._pickCache.gearLen === W.sect.gear.length` instead of a monotonic `W._gearSeq` counter—missing newly forged gear whenever 1 item was sold and 1 item was forged in the same 3-hour window!

2.4 Disciples Spend CP to Buy & Equip `dur = 0` Broken Gear From the Sect Shop (`cpeco.js:288, 338–355`)!

- Neither `CPE.pickPurchase` (`line 288`) nor `CPE.buy` (`line 338`) filters out `gr.broken || gr.retired || (gr.procedural && gr.dur <= 0)`! An

unequipped novice (`cs === 0`) sees a broken weapon's `0.12×` residual score (`18 > 8`), **spends CP to buy the broken `0`-stat item, and equips it into combat**! Filter `!M.gearUsable(n, gr)` out of `CPE.pickPurchase` (unless the buyer owns a repair stone and intends to repair it) and auto-repair armory gear via the `forge` officer!

2.5 Second Herb-Pool Pollution Site (HERB_POOL in `js/cpeco.js:138`)!

- Just like `HERBS`/`ORES` in `economy.js:354–355` (`04C`), `cpeco.js:138` (`A.forage` — เก็บสมุนไพรหุบเขา) pollutes `HERB_POOL[1]` (`3/5 = 60%`) and `HERB_POOL[3]` (`3/6 = 50%`) with **Weasel Fur (`MAT_BRUSH_SPIRIT_WEASEL_HAIR_1`), Cinnabar Mineral Ore (`MAT_BRUSH_STARLIGHT_CINNABAR_3`), Python Tendons, and Iron Silkworm Thread**—slashing real medicinal herb gathering in half! Clean `cpeco.js:138` alongside `economy.js:354–355`.

---

3. `[EQ-SYS-03]` Exile / Betrayal Relic Black-Hole (`visitors_clues.js:1074` & `sim.js:SIM.retire`)

Inspect `VC.syncUniqueArtifacts` (`js/visitors_clues.js:1074–1115`):

- If an NPC **dies (`n.dead === true`)**, lines `1076–1086` rescue all `n.equip` and `n.inv.gear` back into `W.sect.gear`.
- **Bug — Exiled / Defected NPCs (`n.exiled === true`, `n.dead === false`)**:
  - `SIM.retire` (`sim.js`) sets `n.exiled = true` without transferring `n.equip` or `n.inv.gear`!
  - Meanwhile, `VC.syncUniqueArtifacts` (`line 1074`), `isTargetAlreadyClaimed` (`line 1192`), and `findCurrentHolderOfUnique` (`line 1212`) check **`if (n.dead) continue;` WITHOUT checking `if (n.exiled)`**!
  - **Consequence**: If a disciple holding `LEG_WPN_YITIAN_SWORD` or `LEG_ARM_CELESTIAL_CRANE_ROBE` is exiled or defects, **the Unique Legendary Relic stays frozen on the inactive `exiled` record forever—permanently marked `'claimed'` so no Visitor Clue can ever spawn it again, yet unreachable by anyone in the world**!
- **Emergent Wuxia Solution**:
  1. **Formal Sect Expulsion (`TRIBUNAL` / peaceful retirement)**: Confiscate/return Sect-bound & Unique Relics into `W.sect.gear` (or bequeath to successor) before setting `n.exiled = true`.
  2. **Violent Betrayal / Defection Escape**: If a traitor flees with a Unique Relic, register the relic on the rival sect / rogue renegade in `W.world.uniqueArtifacts` (`ownerKind: 'rogue_defector'`) and **spawn a dynamic Pursuit Clue (ล่าสังหารศิษย์ทรยศทวงคืนศาสตราประจำสำนัก)** so

the sect's Law Enforcers can hunt the traitor down and reclaim the stolen treasure!

---

4. [EQ-SYS-04] World & Rival Sect Equipment Symmetry (foe Units, Tournaments, & Debug/Schema Validator Hardening)

2. **Rival Sect Champions & Jianghu Foes Lack Structured Equipment Metadata (`expedition.js`, `explore.js`, `sect.js`)**:
    - Populate `foe.weaponType`, `foe.armorType`, and (on Named Rival Patriarchs/Champions in `WORLD_MARTIAL_CONFERENCE` / `resolveSectSiege`) real `equip` loadouts so `COMBAT.resolveWeaponTactics`, `EFX_WEAPON_MAGNET_DISARM`, and war-trophy drops reflect what the enemy actually wielded in battle!
2. **Extend `tools/validate_data_schema.js` With 8 New Zero-Tolerance Equipment Gates**:
    - Currently `validate_data_schema.js` passes `0 Errors` despite all the bugs uncovered in `04A–05G` because it only checks shallow field presence! Add these **8 automated invariants**:
        1. Every `ALL_GEAR_TYPES` (`23` types) has both `ITEM_PROC_<TYPE>` in `JD.ITEMS.items` and `REC_PROC_<TYPE>` in **both** `JD.ITEMS.recipes` and `ECO.recipeById` with `proceduralOut: true`.
        2. `JD.ITEMS.recipes.length === Object.keys(ECO.recipeById).length` (`0` orphan recipes invisible to `ECO.recipesFor`!).
        3. Every `preferredSubs` entry on `VC.LEGENDARY_EQUIPMENT` and every key in `M.SUB_EQUIP_BONUSES` resolves to a **live** subpath in `CULT.PATHS` (`0` stale pre-alias IDs!).
        4. Every `VC.LEGENDARY_EQUIPMENT` item (`27/27`, including all `6` `v2.0.0` weapons!) has at least 1 reachable template in `VC.CLUE_TEMPLATES`.
        5. Zero raw Hanzi inside `.nameTh` strings across `JD.ITEMS` and `VC.LEGENDARY_EQUIPMENT`.
        6. Every status key referenced in `effects.params.targetStatus` / `selfStatus` / `grantBuff` (`tempoBreak`, `contractDefault`, `gravityCrush`, `meridianSeal`, `ironSkin`, `focusUp`, etc.) has an active stat/combat handler in `STATS` or `COMBAT`.
        7. `ST.gearStats(item)` before and after `JSON.parse(JSON.stringify(item))` produces **byte-identical** stat objects across `q = 0.70 .. 1.85` (`1★..3★`) and after `socketElementalRunes`.

8. Newly minted gear from every source (`mintGear`, `rollProceduralGear`, `EXT.tradeSect`, `CONTENT`, `fishing`, `VC`) carries `listedAtDay === W.day`.

---

5. Master Prioritized Execution Roadmap for GPT & Gemini (`P0` $\to$ `P3` Across `04A–05G`)

Phase	Priority	Target Files	Exact Deliverables
Phase 1	**P0 — Critical Data/Stat Loss & Instant Deletion**	`economy.js`, `external_market.js`, `stats.js`, `martial.js`, `visitors_clues.js`	1) Stamp `listedAtDay` in `mintGear`/`rollProceduralGear` & exempt relics/spirits from `sellStaleSectGear` (`04E/05G`). 2) Consolidate the 4-layer `mintGear` onion & preserve `baseSt` (`source.st` + `applyArmorArtisan`) across rune-socketing and `3★` Save/Load (`qRatio` up to `1.40`) (`05A/05D/05G`). 3) Fix `XIAOLI_HEART_DAGGER` zero-Qi `SHADOW_SAMURAI` lockout (`04A`) & fix `M.gearType` alias normalization (`HEAVY_ARMOR`, `LIGHT_ARMOR`, `SHOES`, `PENDANT`,

Phase	Priority	Target Files	Exact Deliverables
			ORNAMENT) (05A/05B).
Phase 2	**P1 — Dead Combat Procs, Inert Debuffs & Copy-Paste Relics**	visitors_clues.js, stats.js, combat.js, economy.js	1) Split runWeaponOnStruckProcs so DAGOU_STAFF checks melee range and all 3 Legendary Armors + 4 Legendary Accessories execute their true authored effects (including on heal/support/command casts!) (04A/05A/05B). 2) Wire all 7 inert v2.0.0 debuffs (tempoBreak, contractDefault, gravityCrush, meridianSeal, etc.) in stats.js & combat.js (04D). 3) Gate recordWeaponKill (EQ-02), reset adaptArmorToElement (E02) per fight, fix symmetric evaluatePairedItems (E03), and protect priceless stock in socketElementalRunes (04E/05A/05D).

Phase	Priority	Target Files	Exact Deliverables
Phase 3	**P2 — Procedural Engine, Crafting Ecology & Tool/Talisman Loops**	economy.js, cpeco.js, data_items.js, data_martial.js, expedition.js, cultivation.js	1) Sync all 23 REC_PROC_* & 475 expansion recipes into JD.ITEMS.recipes; merge MAT_V230_* materialMods; remove resonanceSetId = matched[0] hijacking; replace harsh it.dur / 100 curve (05F). 2) Clean animal furs/tendons/ores out of HERBS, ORES, and HERB_POOL (04C/05G). 3) Wire TOOLs, Spatial Pouches, and EFX_TRIBULATION_SHIELD from equip.ACCESSORY + inv.gear utility (05B/05C/05E), expend TALISMAN charges (05C), and reconcile combatMult/defMult (05E).
Phase 4	**P3 — Emergent NPC Autonomy, Clues, Lineage Gifts & Schema Gates**	martial.js, cpeco.js, sect.js, cognition.js, explore.js, visitors_clues.js,	1) Author the 6 missing Legendary Weapon Clue Templates (ROSARY, CLAW, NEEDLE, FAN, ASTROLABE, ABACUS) + fix CONDOR_BOW trial

Phase	Priority	Target Files	Exact Deliverables
		validate_data_schema.js	BOW check (04B–04D). 2) Update M.PROFILES & SUB_EQUIP_BONUSES (GOLDEN_NEEDLE_SAINT $\to$ NEEDLE, KARMA_SAMSARA $\to$ ROSARY, SONIC_HARMONY $\to$ ZITHER, INK_LITERATI $\to$ BRUSH, STRATEGY $\to$ FAN/ABACUS) & M.ALL_WEAPON_SKILL_KEYS in sect.js/cognition.js. 3) Route 'gift-worthy' gear to real NPC gifting, add A.personalForge, fix needSummary under-tiered Elder demand, handle exiled relic recovery, and enforce the 8 new validate_data_schema.js gates (05E/05G).